

Machine Learning Based Knowledge Level Classification and Summarization of Educational Video

Shashank P Mane

Department of Electronics and Communication Engineering
KLE Technological University
Hubballi, India
shashankmane2004@gmail.com

Swaroop Prakash Naik

Department of Electronics and Communication Engineering
KLE Technological University
Hubballi, India
swaroopnaik2108@gmail.com

Nandana S Korigoudar

Department of Electronics and Communication Engineering
KLE Technological University
Hubballi, India
nandusk29@gmail.com

Kaushik Mallibhat

Department of Electronics and Communication Engineering
KLE Technological University
Hubballi, India
kaushik@kletech.ac.in

Abstract—Educational videos play a major role in modern learning environments and become a primary medium for knowledge. However, most existing video analysis systems are limited to basic tasks such as summarization or basic segmentation and fail to capture the type of knowledge conveyed in instructional content. Manual evaluation of such content is time-consuming, subjective, and difficult to scale. This limitation makes it difficult for learners and educators to understand the cognitive value of video content and to use videos effectively for structured learning. In order to address this problem, this work proposes an intelligent machine learning based framework that performs educational video summarization combined with knowledge-level tagging. Experimental results show that the proposed system accurately identifies different knowledge levels and produces concise summaries that reflect the cognitive intent of the video content. The proposed framework fills an important research gap and offers a practical tool for enhanced learning analytics, personalized education, and intelligent educational video retrieval.

Keywords—Educational Video Analysis, Knowledge Level Classification, Machine Learning, Natural Language Processing, Video Summarization, Automatic Speech Recognition, knowledge level tagging, Fine tuned BERT, LED for summarization.

I. INTRODUCTION

In today's world, educational videos have become one of the most important forms of content or information delivery in online education field. Some of the platforms like edX, YouTube, and some institute academics video have covered subject related topics from science technology to humanities and arts. The videos can contain different type of knowledge (Factual, Conceptual, Metacognitive, Procedural), and these educational videos or informational videos are designed to enhance understanding the concept by visual demonstrating, conceptual explanations and reflective discussions and step by step procedure, get to know how to learn better.

A. Need for project

With the exponential growth of educational video-material the task of analysing and underlying knowledge types in the video have become increasingly complex. Educators and learners find it difficult to evaluate and instructional depth of the videos, and to know the word knowledge level of the content. So there is a need of summarize of video content and classify it based on the type of knowledge it provides. Students are increasingly depended on most multimedia content to get the knowledge for understanding and skill development. These videos teach many things related on basic science concepts and technical steps and even some ideas but some of the videos are long and detailed, learners often struggle, to quickly find the exact type of information they need.

For example: One student may want a brief summary of the main concept, while another may need a more detailed, step-by-step summary of the procedure. Teachers may want to check whether a video focuses on facts concepts procedures or meta cognitive thinking.

B. Detailed problem statement

Educational video lectures contain a rich mixture of factual, conceptual, procedural and metacognitive knowledge. The current e-learning platform do not have smart tools to analyse videos in detailed level so there is need for classifying video content based on the different types of knowledge. Showing which part of the video match different learning of knowledge levels and reduces the human work. The system can also help by creating some short and meaning summaries and helping for user to discover important information.

C. Background information:

- Factual knowledge: It contains some basic information terminologies and definition.

- Conceptual knowledge: It having some principles and theories and some ideas
- Procedural knowledge: In this knowledge level it contains step by step algorithm or methods or skills.
- Metacognitive knowledge: Awareness and reflection on one's learning along with applications of learning and strategies.
- Some of the terminologies used like Automatic Speech Recognition (It is the process which converting audio to the text (using Wisper)) and second thing is , Knowledge Level Classification (By (fine tuned BERT) assigning the text-segments it defined knowledge categories) and next Summarization (A Natural Language Processing (NLP) task it condenses or shorting long text into concise (LED))

D. Objectives of the proposed work

- Implement a knowledge classification that categorizes the content into knowledge dimension (Factual, Conceptual, Procedural, and Metacognitive).
- To build a machine learning-based model to perform a concise and easy summarization of the educational video transcripts.

II. LITERATURE WORK

The reviewed set of research papers published in between the year 2001 and 2025 shows the gradual evolution of research in educational frameworks and multimodal learning, video summarization. Earlier contributions in this area were largely foundational and theoretical in nature. Works such as the revised Bloom's Taxonomy has given the conceptual basis for understanding learning and knowledge levels, which further influenced educational data analysis research [1], [14]. Similarly, early video summarization efforts focused on methodological development rather than large scale empirical evaluation [5].

From 2020 onwards, the increase in research activity was noticeable as there was rise in peer reviewed journal and conference publications. Studies during this period began leveraging advances in deep learning and multimodal feature extraction to analyse educational videos more effectively [2], [4], [8]. The year 2021 represents a major phase of development, as various influential works introduced deep neural networks, transformer based architectures and audio-visual fusion techniques for video summarization and representation learning [4], [7], [8], [13].

In the recent years (2023–2024), the literature shows a strong dominance of journal publications, reflecting that the field has matured and moved towards comprehensive surveys, systematic studies and refined methodologies [6], [10], [11]. These works emphasized interpretability, robustness and broader applicability of video summarization and learning analytics techniques. The most recent paper in 2025 further confirms that research is in progress, particularly in the area of Educational video analytics and cognitive modelling with a

focus on understanding learning complexity and engagement [9].

Overall, the reviewed papers showcases a clear transition from early conceptual and exploratory studies to advanced, data driven and cognitively informed research in the 2020s. The trend also indicates an increasing preference for journal based publications for presenting mature and validated research, while conference papers continue to play significant role in introducing emerging ideas and novel approaches in video understanding and educational intelligence [4], [6], [13]. Now lets understand by synthesis diagram which help to know classification of paper with there working and its models in fig1.

A. Gaps in current research and how this work addresses them

- Most existing video models only classify or summarize videos. They do not tell which part is factual, conceptual, procedural, or metacognitive.
- Many models use only video frames or text transcript.
- Video summarization models take a long videos. Select important frames or segments ant then create a short summary (keyframes or short clips) But they usually do not explain why those frames were selected.
- Most existing video summarization models are trained on entertainment-oriented videos such as movies and sports, and therefore fail to capture the semantic and pedagogical importance required for summarizing educational videos.
- Summarization, classification, and knowledge tagging are separate as Existing research handles these tasks individually. There is no single system that does all three together. • Almost no research uses Bloom's levels for labelling educational video content.

How the proposed work solves these gaps:

- The model tags each video segment as factual, conceptual, procedural, metacognitive. This fills the biggest gap in previous research.
- The model use video to text conversion as the system uses spoken audio, transcript text. This improves accuracy and gives a complete understanding of each segment.
- The model understands explanations, definitions, examples, problem-solving, reasoning steps. This solves the issue of general-purpose models failing on educational content.
- The model highlights key frames, important words, important explanation. And it solves "black-box" problem.
- The combined tasks in the system performs video summarization, video knowledge-level tagging. This unified system is not available in current research.

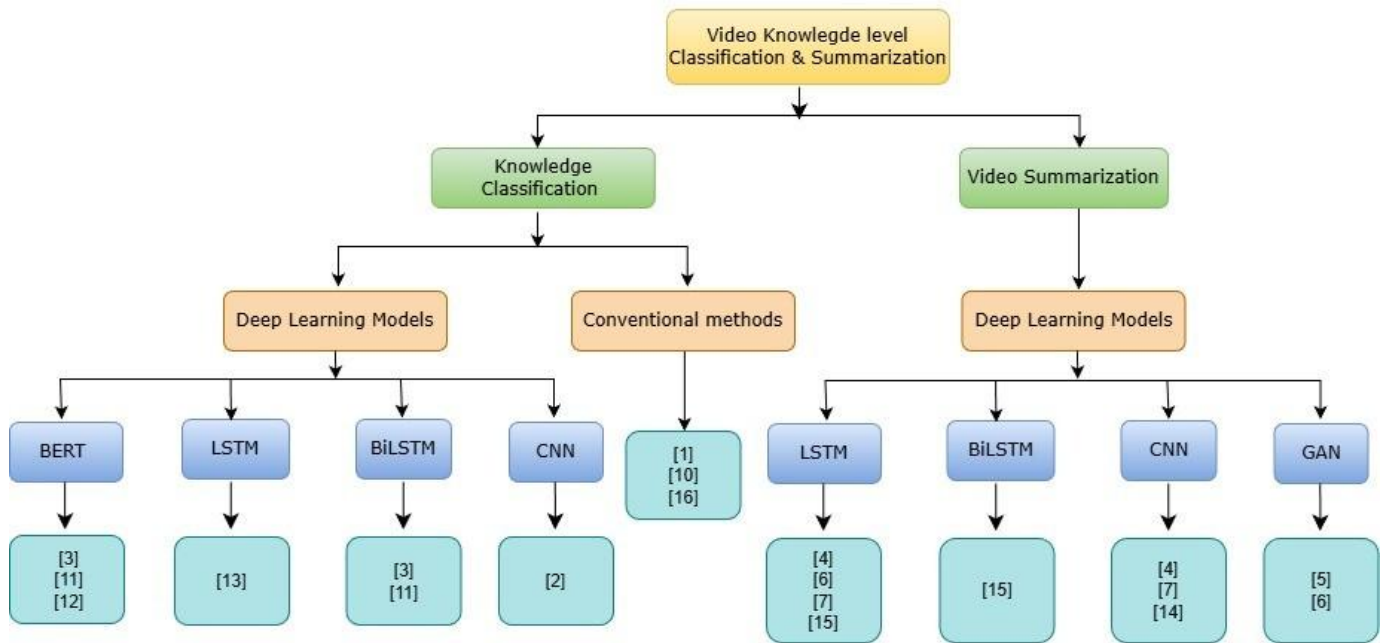


Fig. 1. Synthesis diagram representing models used in various research papers towards Knowledge level classification and summarization applications

B. Summary of related review

knowledge-level tagging, and a unified system combining summarization, classification, and cognitive analysis

In recent years, the research in the area of educational video analysis has grown a lot. Earlier studies focused mainly on understanding structure of learning. These works given the theoretical foundation for how learning content should be understood and organized.

In later years the research moved towards the video classification and video summarization, where the main aim was to automatically identify significant parts of a video or group videos based on the topics. These studies used deep learning, computer vision and audio-text processing for detection of key frames, videos segment, and make summaries. However, most of these approaches were designed for general videos like sports, movies, random YouTube clips and not specifically for educational content. Multimodal learning is introduced more in recent studies, where video, audio and text are combined together to understand content more deeply. Better features from videos are extracted using transformer-based models and large deep-learning networks. These models improved understanding and accuracy but still do not fully consider the learning purpose, teaching structure, or cognitive levels in educational videos.

Among all the related research, one common limitation is that the systems focus mainly on the technical aspects such as summarizing frames or detecting scenes but they do not connect the content to the learning outcomes. Only very few works consider how the information in a video fits into levels like factual, procedural, conceptual or metacognitive knowledge. In summary, prior studies give strong foundations in learning theories, video processing, and automatic summarization. But they lack in educational focus,

III. PROPOSED WORK

Traditional educational content analysis depends on manual labelling and human judgment. This process is slow, inconsistent, time consuming and its difficult to handle large number of educational videos. To solve these problems, the proposed machine learning based model analyse educational videos, extracts important information and summarizes and classifies the content based on learning taxonomy(knowledge level). These system identifies 4 knowledge levels factual, conceptual, procedural, metacognitive.

A. Presented approach

The main aim of this system is to classify educational video content according to knowledge levels-factual, conceptual, procedural, metacognitive and to generate a concise and easy summary of the video. The complete workflow integrates speech recognition, natural language understanding, machine learning, and text summarization. The model processes video input, convert speech to text, segments the transcript into timealigned sentences, and applies fine-tuned BERT for knowledge classification, and summarization of complete transcript, and the final output is knowledge-wise timestamped of video and concise and short summary. The major stages are illustrated as shown.

1.Video Input and Audio Extraction: The system starts with an educational video in .mp4 format. The video stream is processed using tool FFmpeg , which extract the audio track and convert it into a clear and clean standardized waveform.

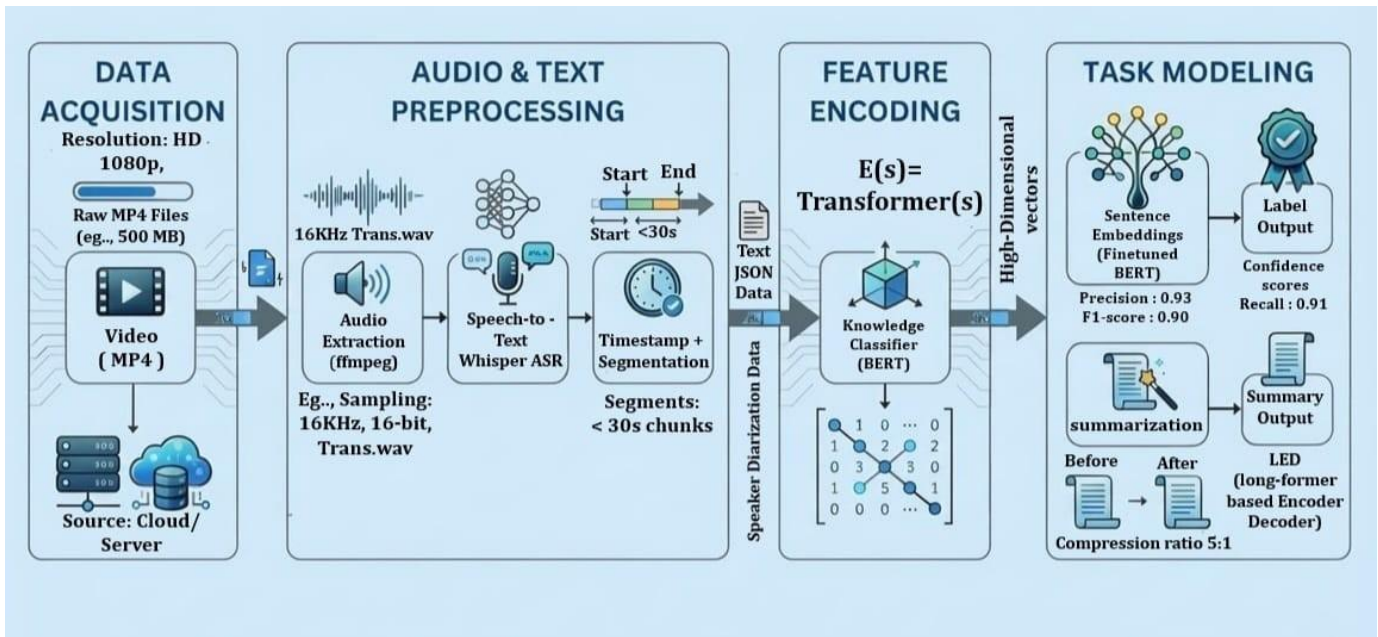


Fig. 2. System architecture for knowledge level classification of a video

Then the extracted audio act as the input to the speech recognition module.

2.Speech-to-Text Transcription using Whisper ASR: Now the extracted audio is processed by using Open AI’s Whisper Automatic Speech Recognition (ASR) model. The whisper is a pre-trained, end-to-end neural model designed to convert speech into text while maintaining high accuracy, whisper ensures high quality transcription even with accents and noise. Whisper first performs feature extraction, where it converts raw audio waves into a Mel-spectrogram representation—a visual form of sound intensity over time and frequency that the model can interpret. The spectrogram is then passed through a deep encoder-decoder neural network which analyses audio features and predicts corresponding text sequences. Long segments from whisper are split into sentences. Each sentence inherits estimated start and end timestamped.

Then the preprocessing stage plays a crucial role since the initial transcript may contain filler words such as “uh”, “umm”, “aaaaahhhh” and other speech disfluencies, these noninformative elements are removed to reduce linguistic noise. The text is then normalized by eliminating extra spaces and formatting inconsistencies to ensure structural clarity. Following this, the continuous transcript is segmented into meaningful sentence-level units while preserving timestamp information obtained from the transcription stage and this ensures that the input to the BERT-based classifier is clean, semantically coherent, and properly structured, thereby improving the overall accuracy and reliability of knowledge-level classification.

The output of this stage is a time-stamped transcript, where each line of text corresponds to the specific time interval

within the video. This enables time-aligned text analysis for subsequent classification.

3.Knowledge-Level Classification using Fine-tuned BERT:

BERT(Bidirectional Encoder Representations from Transformers) developed by Google, is a pre-trained transformer model that understands the semantic meaning and context of human language. It understands context from both left and right direction, it performs well on sentence level classification tasks.

After obtaining the transcript, each sentence or segment is analysed to determine its knowledge type using a finetuned BERT-based multi-class classification model. Initially, the text is tokenized using BERT’s tokenizer and converted into numerical input representations that preserve contextual relationships between words. These tokenized inputs are passed through multiple transformer encoder layers, where self-attention mechanisms capture deep semantic and contextual dependencies within the sentence. The final hidden representation of the special classification token ([CLS]) is extracted and fed into a fully connected classification layer. A softmax function then computes probability scores across the four knowledge categories—Factual, Conceptual, Procedural, and Metacognitive. The category with the highest probability is selected as the predicted knowledge type. Finally, the predicted label is combined with the corresponding timestamp to construct a structured knowledge timeline of the video. For example:

- “Newton’s first law states that...” → Factual
- “This law explains how inertia works...” → Conceptual
- “To find the force, multiply mass by acceleration...” → Procedural

- “When solving problems, always check if friction is considered...” → Metacognitive

The output is a labelled transcript indicating, for instance, that from 0:00 to 1:30 minutes the content is factual, from 1:30 to 4:15 minutes procedural, and so on.

Training phase: Training BERT for knowledge classification task, involves fine-tuning of pre-trained transformer on a labelled educational dataset. Fine-tuning allows the model to adapt its pre-trained language understanding to the specific task of Bloom’s taxonomy-based knowledge classification. As a result, the model learns to distinguish semantic differences between definitions, factual statements, instructional steps, and higher-order reflective content, thereby enabling accurate sentence-level knowledge classification within educational video transcripts.

Training dataset: The dataset primarily contains educational content from class 11 and 12 physics topics, reflecting the instructional language commonly used in academic settings. The dataset contains around 10,000 physics-related sentences. Each sentence is manually labelled into one of four knowledge types: factual, conceptual, procedural, or metacognitive. This labelled dataset serve as supervised ground truth for fine-tuning the BERT-based classification model. By training on domain specific educational sentences, the model learns to distinguish between different types of knowledge representation in instructional content.

Validation phase:

Validation Dataset: A separate dataset of 2,000 physics sentences is used only for testing. This dataset is never used in training. It helps to check how well the trained model performs on new and unseen data.

Validation is performed to evaluate the effectiveness, accuracy, and generalization capability of BERT- based knowledge classification model. Objective of validation is to measure how well model classify unseen educational content.

Using a separate dataset eliminates data leakage and provides a fair evaluation of model performance. Each sentence was fed into fine-tuned BERT model, and predicted were compared with ground-truth labels. Evaluation metrics including accuracy, precision, recall, F1 score used to access performance. The result demonstrated strong classification accuracy and meaningful distinctions across knowledge categories.

Comparison with other models: Several text classification models were evaluated for suitability in this project. Traditional machine learning approaches such as SVM and Logistic Regression rely on bag of words or TF-IDF features and lack of contextual understanding, making them insufficient for capturing the semantic small differences required for knowledge classification. CNN improve performance by learning text patterns, but they mostly focus on short range constraints and may not effectively model deeper semantic relationship. LSTM networks capture the sequential information better than traditional models. However ,they process text in step by step manner and may

struggle with long range bidirectional context, which is crucial for distinguishing between conceptual explanations and factual statements. The proposed fine-tuned BERT model is the most suitable approach for this work because the task requires deep contextual understanding of instructional language rather than simple keyword detection. BERT uses bidirectional self-attention mechanisms to capture rich contextual relationships within sentences. This makes BERT particularly suitable for educational content analysis. Furthermore, fine-tuning BERT on a domain-specific physics dataset allows the model to adapt to educational terminology and instructional patterns.

4.Summarization using LED

- LED (Longformer Encoder-Decoder) is a sequence-to-sequence Transformer designed mainly for long format document understanding and summarization. It extends the Longformer architecture into an encoder–decoder framework so that the very long inputs such as 8K–16K tokens can be summarized in a single pass.
- Initially, the LED encoder takes the entire transcript of the educational video as input and converts it into context representations. It is far different from traditional encoders as it uses sparse attention, where local attention captures the sentence-level details such as procedural and facts steps, while global attention will captures important definitions and concepts in the entire transcript. This helps the encoder to understand long duration videos in a single pass without breaking into segments (chunking), preserving both local meaning as well as Global context.
- After encoding, The LED decoder generates the final summary by processing the encoder’s representations in an autoregressive manner by producing single word at a time. In every step, it attends to the most relevant parts of the encoded transcript while considering previously generated summary tokens. Because the decoder receives a globally coherent representation from the encoder, it produces well-structured, meaningful summaries that accurately showcases the procedures, key concepts and factual information presented in the educational video.

5.Output

The final output of the system consists of:

- 1.A time-stamped knowledge classification report:
- e.g: 0:00 – 1:30 → Factual ; 1:30 – 4:15 → Procedural ; 4:15 – 5:00 → Conceptual.
- Percentage distribution of knowledge types across the entire video, helping educators know the knowledge level of content.
- This ensures accurate, time-aligned labelling of educational content and generates high-quality summaries, reducing manual effort and enabling large-scale analysis of instructional videos.

IV. RESULTS AND ANALYSIS

This section presents the experimental result obtained from the proposed system for summarization and knowledge level classification of the video ,the analysis includes evaluation of transcript quality, classifier performance, validation mechanism and summarized quality, and followed by performance matrix, visualization of solutions, interaction of outcomes and discussions of limitations.

A. Presentation of results

After the raw video is processed, the audio is converted to text using ASR system. The transcript is then divided into a smaller segments based on natural flow of speech, by natural pauses and structures of the speaker's narration then corresponds to a meaningful spoken idea. Once the transcript is segmented , then each segment is analysed and classified into one of the four knowledge type along with start and end timestamp for every segment, making it possible to understand not only what kind of knowledge is present but also when it appears in the video. This classification enables us to understand what type of knowledge is being delivered throughout the video.

After classifying all segments, The model calculate how much total time was devoted to each knowledge category (total duration of each knowledge type by using the timestamps). This determines how much proportion of the video contains factual explanation(more definition-focused), conceptual understanding(understanding-driven), procedural steps(step-by-step oriented) and metacognitive content(reflective/ strategy based/ application based). The resulting distribution is then visualized in the form of a bar graph, showing the proportion of each knowledge level in the entire video.

TABLE I
TIME STAMPED KNOWLEDGE CLASSIFICATION ACROSS THE
COMPLETE VIDEO ON NEWTON'S LAWS OF MOTION

Index	Start Time (s)	End Time (s)	Type
0	0.00	23.64	Metacognitive
1	23.64	104.44	Factual
2	104.44	139.24	Procedural
3	139.24	161.12	Conceptual
4	161.12	169.16	Metacognitive
5	169.16	180.56	Procedural
6	180.56	186.88	Metacognitive
7	186.88	196.40	Procedural
8	196.40	244.96	Factual
9	244.96	257.92	Procedural
10	257.92	270.96	Metacognitive

The proposed system utilizes the BERT-transformer-based bidirectional encoder that captures rich contextual representations from textual data. The model is fine-tuned for a single label multi-class classification task with four output categories: Factual, Conceptual, Procedural, and Metacognitive. A supervised learning approach is employed using a domain-specific annotated dataset. For fine-tuning, the

learning rate is set to $2e-5$, with a batch size of 8, and the model is trained for 3 epochs. The maximum input sequence length is limited to 128 tokens to balance contextual coverage and computational efficiency. Training is performed on Google Colab using an T4 GPU, which accelerates the optimization process and reduces overall training time.

Knowledge Type Distribution (%):

```
type
Conceptual      8.07
Factual         47.74
Metacognitive   18.84
Procedural      25.35
Name: duration, dtype: float64
```

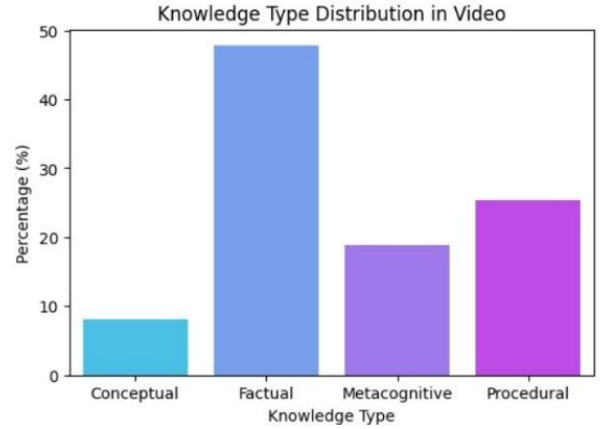


Fig. 3. Knowledge Distribution across the complete video on Newton's Laws of Motion

In some cases, differences in knowledge level classification occurs between the proposed machine learning model and other model or transformer (like GPT/ Open AI) do not indicate an error in the model. Instead, they occur due to fundamental differences training sources, variations in phrases, differences in sentence interpretation, contextual meaning, overlapping meaning between contextual that may lead to different meaning or understanding. The machine learning model learns statistical patterns strictly from the labelled training dataset provided and follows the annotation logic embedded within that dataset. In contrast, other models may work as an independent, human like evaluator that interprets sentences using extensive world knowledge or may perform semantic reasoning in a human like manner and flexible reasoning aligned with its internal understanding of Bloom's taxonomy without being constrained by the project specific dataset.

From a research perspective, this phenomenon is referred to as inter annotator disagreement, where different annotator (trained ML model/GPT) assign different but valid labels to the same input. This behaviour is well documented in natural language processing tasks involving cognitive and educational taxonomies. Therefore, agreement between the ML model and GPT indicates high confidence predictions, while

disagreement signifies ambiguity, multi-level cognition or borderline classification rather than model failure.

In addition evaluation is performed for classification model using standard machine-learning metrics. Standard evaluation metrics were computed, including accuracy, precision, recall, and F1-score. The results shows how strong performance achieved across all four knowledge categories.

TABLE II
PERFORMANCE ANALYSIS OF THE FINE-TUNED BERT MODEL

Class	Precision	Recall	F1-Score	Support
Factual	0.64	0.97	0.77	509
Conceptual	0.95	0.61	0.75	504
Procedural	0.98	0.78	0.87	532
Metacognitive	0.93	0.98	0.96	555
Accuracy	0.84			2100
Macro Avg	0.88	0.84	0.84	2100
Weighted Avg	0.88	0.84	0.84	2100

Human evaluation: To further assess the reliability and practical usability of the proposed knowledge classification, a human validation study is conducted. Human evaluation is important because educational content is often complex, and knowledge tagging may involve subjective interpretations that automated system may struggle to capture. The main purpose of human evaluation is to measure agreement between human expects and the model, to identify categories where the model performs strongly or weakly, then to evaluate real-world applicability.

The sentences from processed video transcript was selected for human evaluation. To ensure fairness, the sentences are randomly selected for evaluation and based on meaning of four knowledge level the sentences are divided. Evaluator did not see the model predicted to avoid bias. The human validation study demonstrated that the proposed knowledge classification system performs consistently and reliably when compared with human judgment. The model achieved strong alignment with expert evaluator. Sometimes disagreement occurs between two knowledge level because the statement can be understand in both way like conceptual statement includes factual definitions. Some sentences may belong to multiple knowledge categories simultaneously, human judgments may vary due to subjective interpretation.

The comparison confirms that, the model is capable of performing classification at a level comparable to trained human evaluators. Automated knowledge classification is feasible and accurate enough for real-world applications like instructional design analysis, smart video indexing, and educational analytics and Human validation strengthens the scientific credibility of the system by demonstrating external validity.

B. Comparison with previous approaches or benchmarks and interpretation of results and their implications

A review of previous methods and benchmark studies shows that most video summarization research focuses on selecting key frames, identifying important time segments, and creating compact video summaries using machine learning and deep learning techniques. Many studies report better summarization performance by using multiple data modalities and advanced model architectures, such as transformer-based networks and generative models. However, these improvements are mainly due to technical optimization. As a result, most existing video summarization approaches focus on video content structure or perceived importance, while they often ignore the deeper cognitive and educational aspects of the material. But this models take a long videos and give meaningful summary without reducing quality of content. And there is much less work on knowledge level classification but some work has been achieved in classifying sentence into bloom level or bloom taxonomy, and tried to classify question and answers based on bloom level and some tried evaluating cognitive level using bloom level but this model try to classify the video content according to knowledge type with time stamp and tells knowledge level through out video and it is still unexplored. Video-level classification is under-explored as most classify questions or assessment items or design instructional content according to bloom's level.

C. Connecting the results with objectives

To address the gap, the proposed approach presents an integrated framework that performs knowledge-level labelling and video summarization, where the assignment of cognitive levels is formulated as a classification problem. The proposed system assigns suitable cognitive categories to individual video segments, allowing for a more meaningful pedagogical interpretation of educational material that is not supported by earlier methods. The integration of automated summarization with cognitive-level analysis introduces a distinctive capability that is not found in prior studies or available benchmark datasets.

V. CONCLUSION

The results of this project indicate that the proposed pipeline is highly effective for analysing educational videos through knowledge-level classification and abstract summarization. The results show that the system is capable of converting long, unstructured educational or instructional videos into organized knowledge-level segments with classifying them to knowledge level, while also generating helpful summaries. The proposed approach addresses key challenges faced in educational content analysis, including the manual effort required for annotation, understanding required concept and the difficulty of understanding long videos, and the need for consistent classification of instructional material according to learning taxonomies.

TABLE III
COMPARATIVE ANALYSIS OF MODEL PREDICTION AND HUMAN ANNOTATIONS ON FOUR DIFFERENT VIDEOS

Video	Timestamp	Segment	Model	Human	Score
1	0:20–0:22	One is intrinsic semiconductor material.	Factual	Factual	1
1	0:39–0:44	And with semiconductor material if you add impurities...	Conceptual	Conceptual	1
1	1:39–1:41	Missing electron means there is a free hole.	Factual	Conceptual	0
1	2:25–2:28	N type material is having electrons as a majority carrier.	Factual	Factual	1
2	0:04–0:09	Imagine you're sitting on a recliner chair watching...	Metacognitive	Metacognitive	1
2	0:52–0:55	If it's at rest, it will remain at rest.	Metacognitive	Conceptual	0
2	1:05–1:09	If it's at rest and an unbalanced force acts on it...	Conceptual	Conceptual	1
2	1:35–1:40	In simple words, it means that the objects continue...	Factual	Conceptual	0
3	0:07–0:10	Law is based on conservation of energy.	Factual	Factual	1
3	3:49–3:45	If any change of magnetic flux occurs, the coil will...	Conceptual	Conceptual	1
3	4:08–4:14	Ok, now let's move to the second case in which...	Procedural	Procedural	1
3	6:15–6:20	With a force of higher magnitude so that to oppose...	Factual	Cognitive	0
4	0:16–0:25	Now the important question is what happens when...	Metacognitive	Metacognitive	1
4	0:45–0:51	In the first case the object will come to remain at...	Procedural	Conceptual	0
4	2:49–2:54	In simple terms the acceleration is directly proportional...	Factual	Factual	1
4	3:38–3:41	So if the mass of the object is 10 KG and the acceleration...	Factual	Factual	1

Overall, the key findings demonstrate that the designed system is capable of transforming long, unstructured educational videos into organized, organized knowledge-level segments while also generating helpful summaries, making it a valuable tool for both learners and educators.

A. Future works and recommendation

Despite these achievements, there are several opportunities exist for future enhancement. Classification performance could be further improved by expanding and diversifying the labelled dataset, would help increase the classification accuracy and make the model more robust for different subjects, and finetuning transformer-based classifiers specifically on educational could also lead to better performance compared to traditional classifiers. Additionally incorporating Whisper ASR methods which may yield more accurate timestamp alignments which improve segment-level labelling. Using reinforcement learning-based summarization models could produce richer and more context-aware summaries and incorporating multi-stage summarization strategies. A possible extension of this work is to build an interactive learning platform and visualization dashboards for classroom deployment, where students and teachers can upload videos and instantly view knowledge level breakdowns and summaries, which can enhance system usability and support continuous model improvement, enabling broader application in intelligent tutoring systems, learning analytics, and large-scale educational content analysis.

REFERENCES

- [1] Anderson, Lorin W., and David R. Krathwohl. A taxonomy for learning, teaching, and assessing: A revision of Bloom's taxonomy of educational objectives: complete edition. Addison Wesley Longman, Inc., 2001.
- [2] Ghauri, Junaid Ahmed, Sherzod Hakimov, and Ralph Ewerth. "Classification of important segments in educational videos using multimodal features." arXiv preprint arXiv:2010.13626 (2020).
- [3] Li, Hang, Tianlong Xu, Jiliang Tang, and Qingsong Wen. "Knowledge tagging system on math questions via llms with flexible demonstration retriever." arXiv preprint arXiv:2406.13885 (2024).
- [4] Apostolidis, Evlampios, Eleni Adamantidou, Alexandros I. Metsai, Vasileios Mezaris, and Ioannis Patras. "Video summarization using deep neural networks: A survey." Proceedings of the IEEE 109, no. 11 (2021): 1838-1863.
- [5] Lei, Zhuo, Chao Zhang, Qian Zhang, and Guoping Qiu. "FrameRank: a text processing approach to video summarization." In 2019 IEEE International Conference on Multimedia and Expo (ICME), pp. 368373. IEEE, 2019.
- [6] Alaa, Toqa, Ahmad Mongy, Assem Bakr, Mariam Diab, and Walid Gomaa. "Video Summarization Techniques: A Comprehensive Review." arXiv preprint arXiv:2410.04449 (2024).
- [7] Apostolidis, Evlampios, Eleni Adamantidou, Alexandros I. Metsai, Vasileios Mezaris, and Ioannis Patras. "Video summarization using deep neural networks: A survey." Proceedings of the IEEE 109, no. 11 (2021): 1838-1863.
- [8] Akbari, Hassan, Liangzhe Yuan, Rui Qian, Wei-Hong Chuang, Shih-Fu Chang, Yin Cui, and Boqing Gong. "Vatt: Transformers for multimodal self-supervised learning from raw video, audio and text." Advances in neural information processing systems 34 (2021): 24206-24221.
- [9] Ming, Nong, Sachin Sharma, and Jiho Noh. "YouLeQD: Decoding the Cognitive Complexity of Questions and Engagement in Online Educational Videos from Learners' Perspectives." arXiv preprint arXiv:2501.11712 (2025).
- [10] Navarrete, Evelyn, Andreas Nehring, Sascha Schanze, Ralph Ewerth, and Anett Hoppe. "A closer look into recent video-based learning research: A comprehensive review of video characteristics, tools, technologies, and learning effectiveness." International Journal of Artificial Intelligence in Education (2025): 1-64.
- [11] Yu, Shanshan, Jindian Su, and Da Luo. "Improving bert-based text classification with auxiliary sentence and domain knowledge." IEEE Access 7 (2019): 176600-176612.
- [12] Shaikh, Sarang, Sher Muhammad Daudpotta, and Ali Shariq Imran. "Bloom's learning outcomes' automatic classification using LSTM and pretrained word embeddings." Ieee Access 9 (2021): 117887-117909.
- [13] Zhao, Bin, Maoguo Gong, and Xuelong Li. "Audiovisual video summarization." IEEE Transactions on Neural Networks and Learning Systems 34, no. 8 (2021): 5181-5188.
- [14] Vukic, urica, Sanda Martin' ci' c-Ip' si' c, and Ana Me' strovi' c. "Structural' analysis of factual, conceptual, procedural, and metacognitive knowledge in a multidimensional knowledge network." Complexity 2020, no. 1 (2020): 9407162.